

Normalization In-Class Activity

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Relation

Registration(student_id, student_name, phone, address, major, course_id, course_name, instructor, instructor_type, grade, location, time, session)

Candidate Keys

The only attribute combination that uniquely determines every other attribute in the relation is (student_id, course_id). A student can register for many courses and a course has many students, so neither attribute alone is unique; grade specifically only makes sense for a given student in a given course. No other candidate key exists in the unnormalized relation.

Functional Dependencies

- student_id --> student_name, phone, address, major
- course_id ---> course_name, instructor, location, time, session
- instructor ----> instructor_type
- student_id, course_id ----> grade

student_id and course_id are each partial determinants (they determine attributes using only part of the composite key). instructor ----> instructor_type is a transitive dependency, since instructor_type depends on instructor, which itself depends on course_id rather than on the key directly. Only grade is fully dependent on the whole key (student_id, course_id).

Convert to 2NF

2NF removes partial dependencies attributes that depend on only part of the composite key. student_name, phone, address, and major depend only on student_id, and course_name, instructor, location, time, and session depend only on course_id, so each group is moved into its own relation.

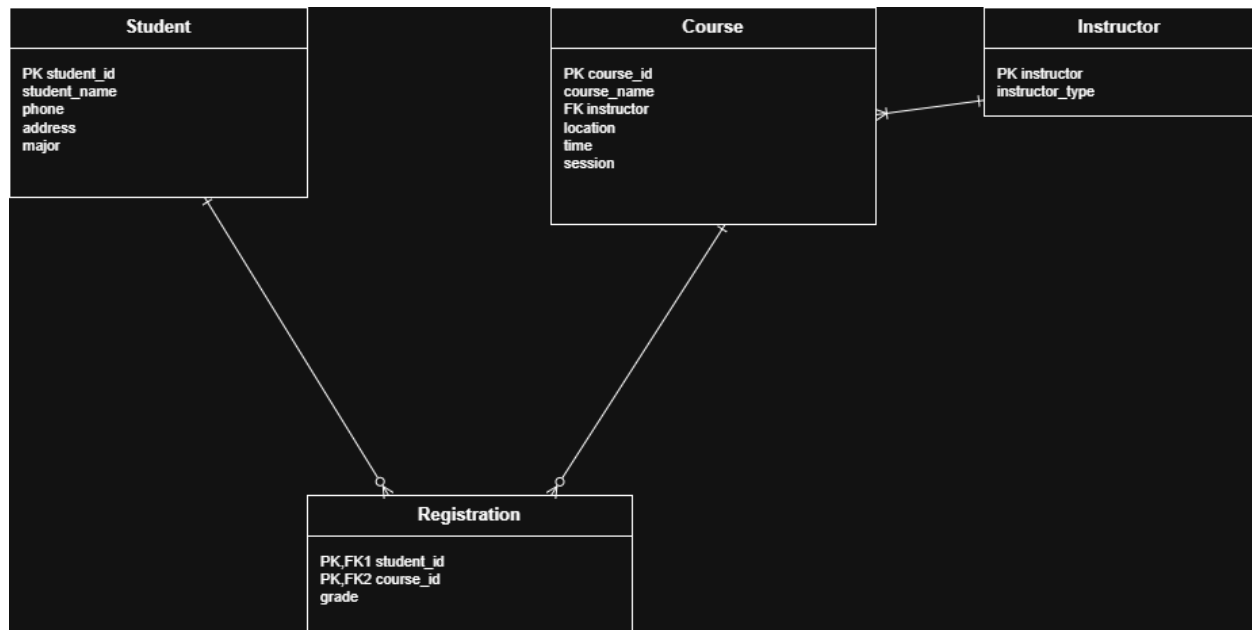
Student	Course	Registration
student_id (PK)	course_id (PK)	student_id (PK, FK)
student_name	course_name	course_id (PK, FK)
phone	instructor	grade
address	instructor_type	
major	location	
	time	
	session	

This is still not 3NF: instructor_type depends on instructor, not directly on course_id, so Course still contains a transitive dependency

Final 3NF relations:

- Student(student_id, student_name, phone, address, major)
- Instructor(instructor, instructor_type)
- Course(course_id, course_name, instructor, location, time, session), FK instructor ---> Instructor
- Registration(student_id, course_id, grade), FK student_id --> Student, FK course_id --> Course

Crow's Foot ERD



SIGN:

-||- one and only one

PK = Primary Key FK = Foreign Key

-o< zero or many

-1< one or many